

Node.js Complete Course (Basic to Advanced)

Module 1: Introduction to Node.js

Lesson 1

- What is Node.js?
 - History of Node.js
 - Why Node.js?
 - Features of Node.js
 - Advantages & Disadvantages
 - Node.js vs Browser JavaScript
 - Node.js vs Other Backend Technologies
 - Node.js Architecture
 - Event-Driven Architecture
 - Non-Blocking I/O
-

Lesson 2

Installation & Setup

- Install Node.js
- Verify Installation
- Install VS Code
- Install Extensions
- Node REPL

Commands

```
node -v
```

```
npm -v
```

Run First Program

```
console.log("Welcome to Node.js");
```

Module 2: Node.js Fundamentals

Lesson 3

- Running JavaScript with Node
- Global Objects
- process Object
- console Object
- __dirname
- __filename

Practice

- Print System Information
 - Display Current Directory
-

Lesson 4

Modules

- CommonJS
- require()
- module.exports
- exports

Example

```
const math = require('./math');
```

Projects

- Calculator Module
 - Utility Module
-

Module 3: NPM (Node Package Manager)

Lesson 5

- What is NPM?

- package.json
- package-lock.json
- Dependencies
- Dev Dependencies

Commands

npm init

npm install

npm uninstall

npm update

Practice

Install

- lodash
 - nodemon
-

Module 4: File System (fs)

Lesson 6

- Read File
- Write File
- Append File
- Rename File
- Delete File

Methods

- readFile()
- writeFile()
- appendFile()
- unlink()

Projects

- Notes Manager
 - Student Records
-

Module 5: Path Module

Lesson 7

Methods

- `join()`
- `basename()`
- `dirname()`
- `extname()`
- `resolve()`

Projects

- File Organizer
-

Module 6: OS Module

Lesson 8

Methods

- `hostname()`
- `platform()`
- `cpus()`
- `freemem()`
- `totalmem()`
- `uptime()`

Projects

- System Information Tool
-

Module 7: Events

Lesson 9

- EventEmitter
- Creating Events
- Listening to Events
- Multiple Events

Projects

- Chat Notification System
-

Module 8: HTTP Module

Lesson 10

Create Server

```
const http = require("http");
```

Methods

- `createServer()`
- `listen()`

Projects

- Hello Server
 - Routing Example
-

Module 9: URL Module

Lesson 11

Topics

- URL Parsing
- Query Parameters
- URL Object

Projects

- Search API

Module 10: Streams

Lesson 12

- Read Stream
- Write Stream
- Pipe
- Buffer

Projects

- File Copy Application
-

Module 11: Express.js Introduction

Lesson 13

- What is Express?
- Installing Express
- Express Server
- Routing

Commands

`npm install express`

Projects

- Student API
-

Module 12: Routing

Lesson 14

- GET
- POST

- PUT
- DELETE
- PATCH

Projects

- CRUD API
-

Module 13: Middleware

Lesson 15

Types

- Application Middleware
- Router Middleware
- Error Middleware
- Third-party Middleware

Projects

- Request Logger
-

Module 14: Static Files

Lesson 16

- Express Static
- Serving HTML
- Serving CSS
- Serving Images

Projects

- Company Website Backend
-

Module 15: Template Engines

Lesson 17

- EJS
- Pug
- Handlebars

Projects

- Employee Portal
-

Module 16: REST API

Lesson 18

- REST Principles
- API Design
- CRUD Operations

Projects

- Student Management API
-

Module 17: MongoDB Integration

Lesson 19

- MongoDB
- Mongoose
- Models
- Schemas

Projects

- Employee Database
-

Module 18: Authentication

Lesson 20

- Register
- Login
- Password Hashing
- bcrypt
- JWT Authentication

Projects

- Login System
-

Module 19: File Upload

Lesson 21

Package

- Multer

Projects

- Profile Picture Upload
 - Resume Upload
-

Module 20: Email

Lesson 22

Package

- Nodemailer

Projects

- OTP Verification
 - Contact Form Email
-

Module 21: Environment Variables

Lesson 23

Package

- dotenv

Topics

- .env
 - Secrets
 - Configuration
-

Module 22: Error Handling

Lesson 24

- try/catch
 - Async Errors
 - Global Error Handler
 - Custom Error Handler
-

Module 23: Async Programming

Lesson 25

- Callback
- Promise
- Async/Await

Projects

- API Calls
 - Database Operations
-

Module 24: Security

Lesson 26

Packages

- helmet
- cors
- express-rate-limit

Topics

- CORS
 - XSS
 - CSRF
 - SQL Injection
 - NoSQL Injection
-

Module 25: Testing

Lesson 27

Tools

- Jest
- Supertest

Projects

- API Testing
-

Module 26: Deployment

Lesson 28

Deploy Applications

- Render
- Railway
- Vercel (Backend Functions)
- VPS Basics

Topics

- Production Environment
 - Build Process
-

Module 27: Performance

Lesson 29

Topics

- Clustering
 - Compression
 - Caching
 - Logging
 - Performance Monitoring
-

Module 28: Final Projects

Lesson 30

Build Complete Real-World Applications

Projects

1. Student Management System
2. Employee Management System
3. Library Management System
4. Hospital Management System
5. E-Commerce Backend
6. Blog API
7. Authentication System
8. File Upload System
9. Chat Backend
10. Online Examination Backend

11. Banking API
 12. Inventory Management System
 13. Hotel Booking Backend
 14. Learning Management System API
 15. Food Delivery Backend
 16. Job Portal Backend
 17. Movie Booking API
 18. Expense Tracker API
 19. School Management System
 20. CRM Backend
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30 Hands-on Practice Programs

1. Hello World
2. System Information App
3. File Reader
4. File Writer
5. Calculator Module
6. HTTP Server
7. Routing Example
8. Express Server
9. CRUD API
10. Middleware Example
11. Static Website Server
12. EJS Project
13. Student REST API
14. Employee CRUD API
15. MongoDB CRUD
16. User Registration
17. Login Authentication
18. JWT Authentication
19. File Upload
20. Email Sender
21. OTP Generator
22. Contact Form Backend
23. Todo API
24. Notes API
25. Blog API
26. Library API
27. Hospital API
28. LMS API
29. E-Commerce Backend
30. Complete Authentication Project

Node.js Interview Questions (Basic to Advanced)

Beginner

1. What is Node.js?
2. What are the advantages of Node.js?
3. What is NPM?
4. What is package.json?
5. What is the Event Loop?
6. What are modules?
7. Difference between `require()` and `import`?
8. What is `__dirname`?
9. Explain synchronous vs asynchronous programming.
10. What is Express.js?

Intermediate

11. What is middleware?
12. Explain routing in Express.
13. What are REST APIs?
14. What is Mongoose?
15. Difference between MongoDB and Mongoose?
16. What is JWT?
17. What is bcrypt?
18. Explain CORS.
19. What is dotenv?
20. Explain file uploads with Multer.

Advanced

21. Explain the Event Loop in detail.
 22. What are Streams?
 23. What are Buffers?
 24. Explain clustering in Node.js.
 25. What is rate limiting?
 26. How do you secure a Node.js application?
 27. Explain `async/await` internally.
 28. How do you handle global errors?
 29. What are worker threads?
 30. How do you optimize Node.js performance?
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20 Real-World Backend Projects

1. Student Management System
 2. Employee Management System
 3. Library Management System
 4. Hospital Management System
 5. Hotel Booking API
 6. Food Delivery Backend
 7. E-Commerce Backend
 8. Banking API
 9. Expense Tracker API
 10. Inventory Management System
 11. Learning Management System (LMS)
 12. CRM Backend
 13. Online Examination System
 14. Blog Management API
 15. Chat Application Backend
 16. Job Portal Backend
 17. Movie Ticket Booking API
 18. Online Voting System
 19. Courier Tracking System
 20. Attendance Management System
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Node.js Cheat Sheet

Node & NPM

```
node -v
npm -v
npm init
npm install
npm install <package-name>
npm uninstall <package-name>
npm update
```

Modules

```
const fs = require("fs");
const path = require("path");
const os = require("os");
const http = require("http");
```

Express

```
npm install express
const express = require("express");
const app = express();
```

```
app.listen(3000);
```

Routing

```
app.get("/", handler);
app.post("/users", handler);
app.put("/users/:id", handler);
app.delete("/users/:id", handler);
```

MongoDB

```
npm install mongoose
```

Authentication

```
npm install bcrypt jsonwebtoken
```

File Upload

```
npm install multer
```

Email

```
npm install nodemailer
```

Environment Variables

```
npm install dotenv
```